Interview brief

Dominion Energy Virginia reliability: what public data shows, and where I'd look first

Independent analysis of public data, for the panel interview on Tuesday 3 November 2026.

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Every figure comes from public data (EIA-861 and PUDL, NOAA Storm Events, EAGLE-I county outages, EIA-930) or Dominion's own publications. No Dominion internal data was used. Items marked [you: ...] are for you to fill in.

The answer first

Frequency, weather and a new load shape

Dominion's reported outage minutes rose from 130 to 133. Rebuilt from public EIA-861 data they are 130.2 and 133.0, so the public data reconciles. The rise came from customers losing power more often, not from longer outages.

+10%

interruptions per customer (SAIFI), 2022 → 2025: 1.199 → 1.320, the highest of 2013–2025

4th of 12

on frequency among twelve peers (SAIFI 2021–25: 1.21 vs median 0.96); restoration time 6th of 12

+39–41%

average load in the hours ending 1–5 a.m., 2019 → 2025, against about +27% for 1–5 p.m. (PJM DOM zone)

The ten-minute talk, slide by slide

#	Slide	Say it in one line
1	Title	Independent analysis of public data; nothing internal.
2	130 → 133: more frequent, not longer	2025 replaced 2022 in the 3-year window; between them SAIFI rose 10% and CAIDI fell 4%.
3	Twelve peers	Mid-pack on restoration time (6th of 12), 4th highest on frequency.
4	Weather	County-days with a storm report had about 12 times the outage customer-hours of calm days (11.9×, 95% CI 7.9–16.9): an association, not a cause.
5	Demand shape	Small hours +39–41%, afternoons about +27%; winter set the peak in 2019, 2022 and 2025.
6	First 90 days	Reconcile, diagnose, prioritise and track; ends in a Power BI programme-tracking report.
7	How it's built	Snowflake RAW → STG → MART, tests, dictionary, decisions; 13 of 13 rebuilt values match the published figures.
8	Close	Frequency first, net of weather, plan for a new load shape.

Say it carefully: the two-sided lines

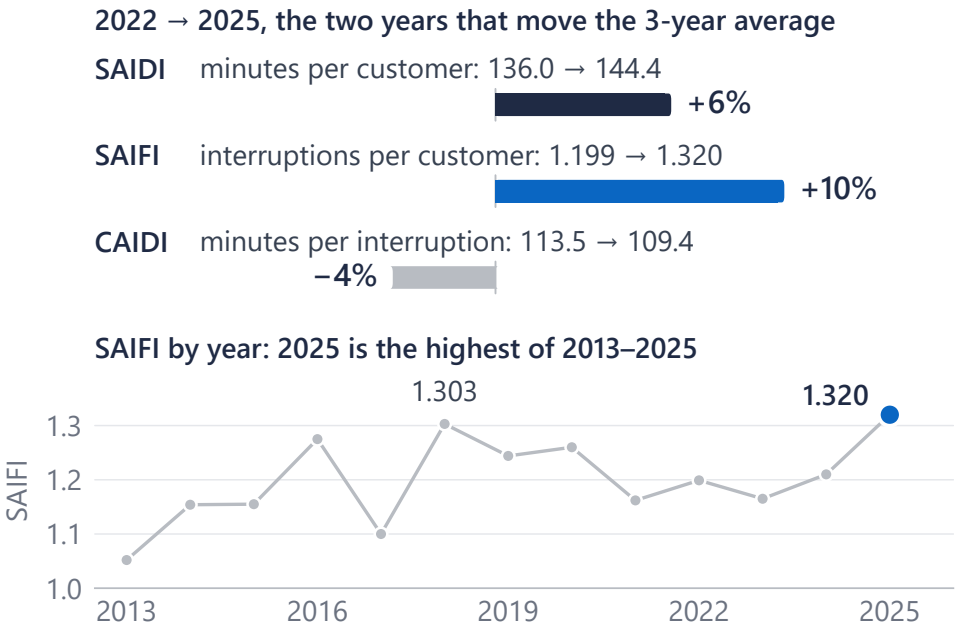
- Measured from 2023 instead of 2022, 79% of the change (in log terms) is still frequency.
- 2025's outage minutes (144.4) are still below 2018 (149.6) and 2019 (149.9).
- Over 2015–2025 no trend is statistically clear (SAIFI p = 0.54, CAIDI p = 0.74, SAIDI p = 0.58): one year is not a trend.
- Adjusted for storm days, there is no clear trend since 2015: about +0.34 minutes a year, 95% range −1.41 to +2.08.

Finding 1

Outage minutes rose from 130 to 133 because interruptions became more frequent, not longer

SAIDI, the average outage minutes per customer, is SAIFI (how often a customer loses power) times CAIDI (how long each interruption lasts). Dominion's 2025 Annual Report gives a 3-year SAIDI, excluding major events, of 133 minutes for 2023–25, up from 130 for 2022–24, and explains it (p. 10): "This increase is primarily due to increased storm activity." The public rebuild gives 130.2 and 133.0.

A 3-year average moves only when a year leaves the window: 2025 (144.4 minutes) replaced 2022 (136.0). Between those two years SAIFI rose 10% and CAIDI fell 4%, so the whole rise is frequency. That is consistent with Dominion's explanation: storm county-days were higher in 2025 than in 2022 (791 against 718; Finding 3). 2025's SAIFI, 1.320, is the highest in EIA's 2013–2025 record (the next highest is 1.303 in 2018).



Source: EIA-861 2025 final and PUDL (EIA-861) 2013–2024, Dominion Energy Virginia (utility 19876), excluding major event days. Reported figures: Dominion Energy 2025 Annual Report, p. 10.

The same answer from either base year

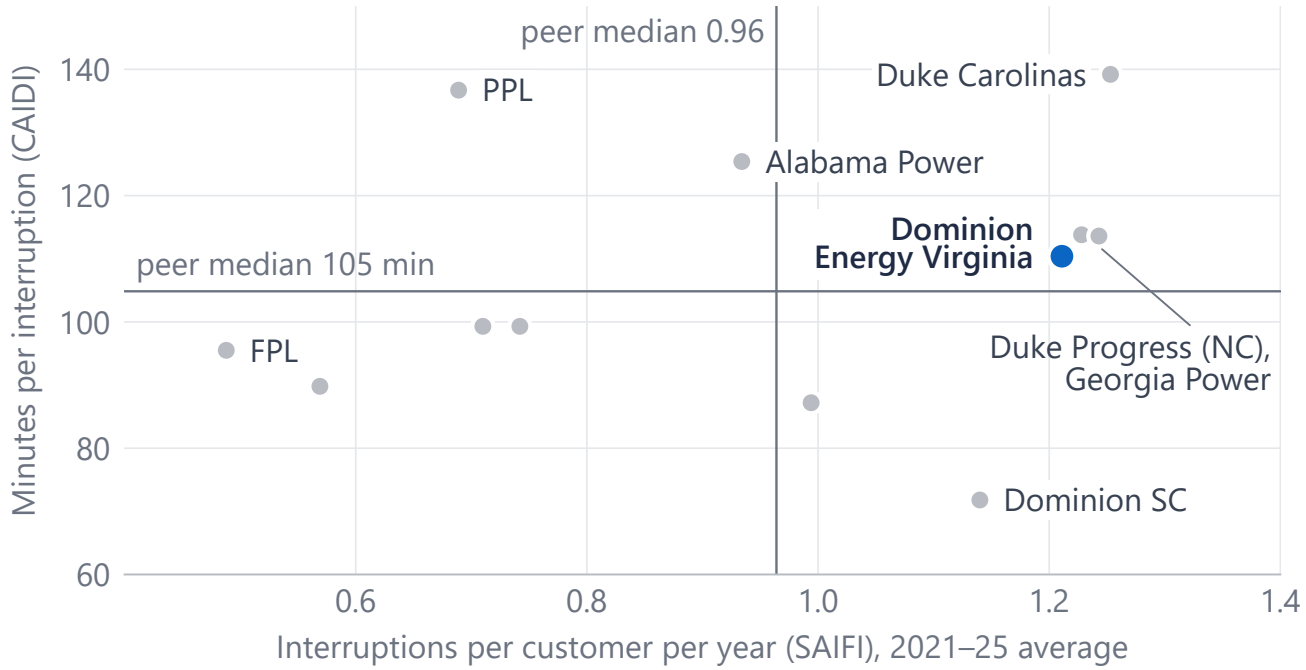
Base year → 2025	SAIDI	SAIFI	CAIDI	Share from frequency
2022 (the year that left the window)	+6.1%	+10.1%	−3.6%	all of it
2023	+17.1%	+13.3%	+3.3%	79%

If asked "so reliability got worse?" "Customers lost power more often in 2025 than in any year of EIA's 2013–2025 record, while restoration time was flat to lower. One year isn't a trend, and 2025's minutes are still below 2018 and 2019, so I'd look at what drove the extra interruptions, by circuit and cause, before calling it a trend."

Finding 2

Among 12 peer utilities, Dominion restores at a mid-pack pace; frequency is the gap

Twelve large investor-owned utilities in PJM and the Southeast, chosen by EIA ID before any result was seen. Averages of 2021–2025, excluding major event days. Dominion's SAIFI of 1.21 is the 4th highest of 12 (peer median 0.96); its CAIDI of 110 minutes is 6th of 12 (median 105). Its nearest neighbours are Duke Energy Progress and Georgia Power. On 3-year SAIDI (2023–25) Dominion is also 4th of 12, at 133.0 minutes.



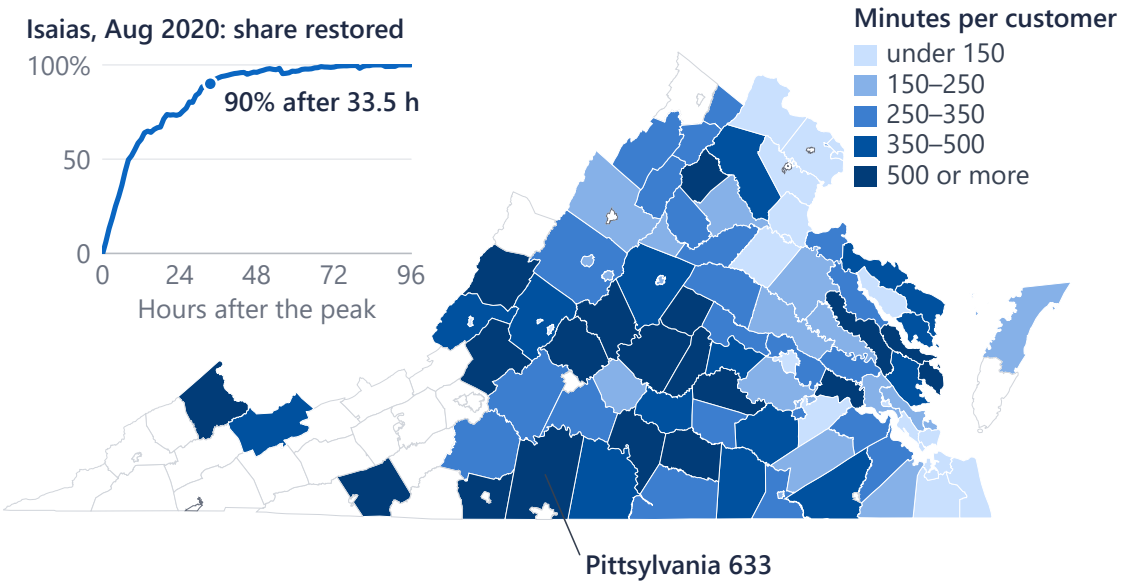
Source: EIA-861 / PUDL 2021–2025, excluding major event days; means of annual values. The full table is in Appendix B.

The caveat to raise yourself: utilities set aside major event days differently. Most of these peers use the IEEE 1366 method; Dominion reports under its own rule, so the ranking is indicative, not exact. The pattern still matches Finding 1: the gap is how often customers lose power, not how fast it comes back.

Finding 3

County-days with a storm report had about 12 times the outage customer-hours of calm days

EAGLE-I, the Department of Energy's county outage record built from utilities' outage maps, gives the customers out every 15 minutes in each county. On the 6,108 county-days from 2015 to 2025 with a NOAA wind, winter or tropical storm report in a county Dominion serves, outage customer-hours per 1,000 customers were 11.9 times those of calm county-days (95% CI 7.9–16.9). It holds on non-major days (9.9×), in Dominion-only counties (19.9×) and in every year. It is an association: storm reports are partly triggered by damage. The map shows where ordinary-day outage minutes are highest: among counties with 20,000 or more customers, from 55 a year (Prince William County) to 633 (Pittsylvania County).



Source: EAGLE-I (ORNL for DOE), customers of all utilities in 99 of the 103 counties where EIA-861 2025 lists Dominion Energy Virginia (four cities left off: unusable data); NOAA Storm Events. Map: outage minutes per customer a year on non-major days, 2015–2024 average. Inset: Isaias, share of the excess outages restored, by hours after the peak.

Two more things to know

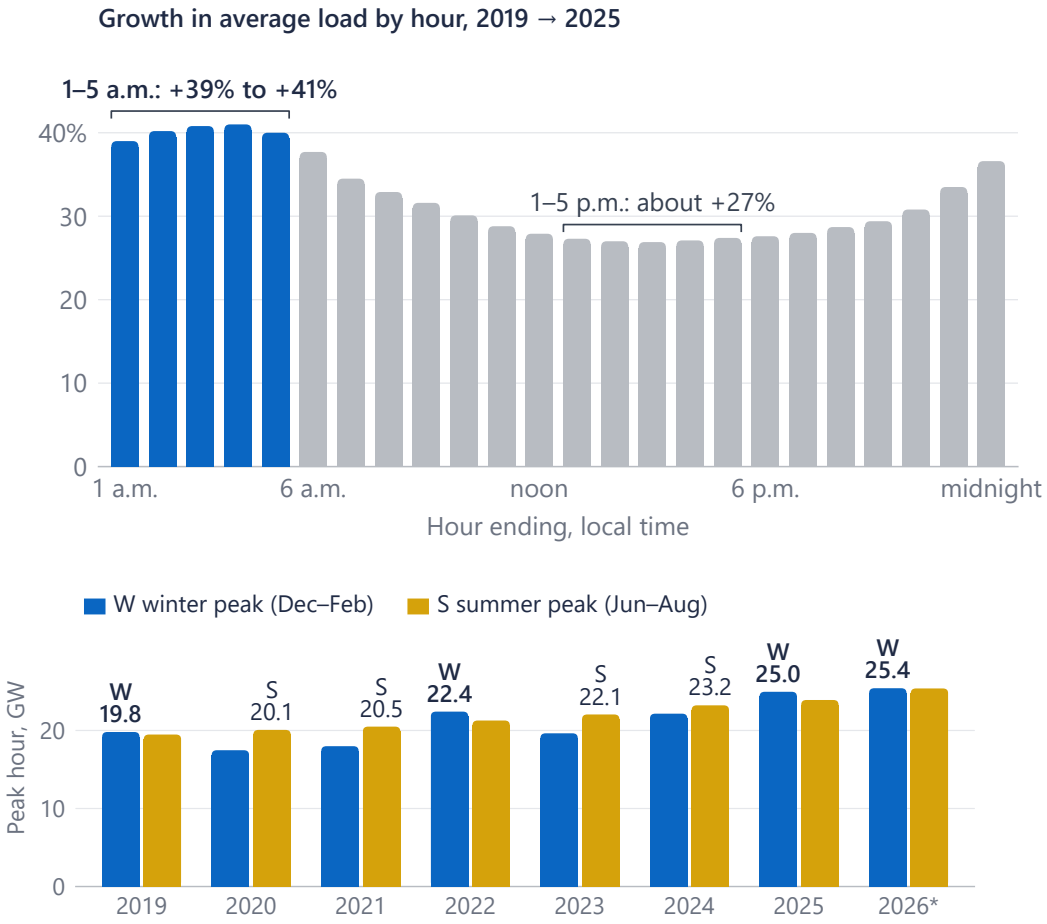
- Year by year, storm county-days and SAIDI excluding major events move together: r = 0.70 over 2015–2024 (n = 10). There were 791 storm county-days in 2025, up from 464 in 2024; a fit made without 2025 gives 145.3 minutes, against 144.4 actual. Adjusted for storm days, there is no clear trend since 2015 (+0.34 minutes a year, 95% range −1.41 to +2.08, p = 0.67).
- Restoration: across 47 events with 50,000 or more customers out, there is no evidence it got faster (hours to 90% restored +4.6% a year after allowing for size, p = 0.07). Isaias: 90% restored 33.5 hours after the peak.

What it cannot say: which circuits, which causes, or whether hardened lines held up. EAGLE-I counts every utility in a county, so it shows where and when, never Dominion's own SAIDI.

Finding 4

Overnight demand grew faster than afternoon demand, and winter rivals summer for the yearly peak

EIA-930 hourly demand for the PJM Dominion (DOM) zone, which is wider than Dominion's retail territory. From 2019 to 2025, average load in the hours ending 1–5 a.m. rose 39–41%, against about 27% in the hours ending 1–5 p.m. Weather does not explain it: holding Virginia's daily heating and cooling degree days and the weekday fixed, average load rose 33% (32% unadjusted). The highest hour of the year rose 26% (19,803 → 24,957 MW) and the lowest 52% (7,167 → 10,903 MW). Winter set the yearly peak in 2019, 2022 and 2025; 2026, to 29 October, is a tie (25,413 MW winter vs 25,384 MW summer).



Source: EIA-930 hourly demand, PJM DOM zone; *2026 to the hour ending midnight, 29 October (winter = January–February only). Winter = December–February, summer = June–August. Weather adjustment: daily load on Virginia degree days and weekday, 2019–2025 (exploratory).

- Of the 50 highest hours in the EIA-930 record from July 2018 to December 2025, 47 fell in 2025.
- Commercial electricity sales in Dominion's Virginia territory rose 65% from 2013 to 2025 (39.08 → 64.62 TWh) on 8% more commercial customers (EIA-861).

Don't say "data centers caused it". Growth around the clock is consistent with large constant loads, but these data cannot identify customers. Say: "less overnight slack, and winter readiness matters as much as summer."

The role, and what I would do in the first 90 days

The posting that most likely matches this interview is **Associate Business Intelligence Analyst / Business Intelligence Analyst** (Richmond, hybrid: three days in the office, two remote). It supports **Grid Resiliency programmes** with Power BI dashboards and reports, and asks for Power BI, SQL, Excel, Snowflake, Power Apps and SharePoint, with reliability metrics as preferred knowledge. [you: confirm the exact title and team with the recruiter]

Grid Resiliency, from Dominion's own description

- Four programmes: mainfeeder hardening, voltage island mitigation, voltage optimisation, and intelligent grid devices (fault location, isolation and restoration). The Strategic Underground Program is separate.
- Mainfeeder hardening selects feeders with at least 500 customers whose average SAIDI is more than 2 minutes worse than the rest, and with above-average outages over the last five years.
- Dominion reports a 32% reduction in outage minutes on feeders hardened for at least 12 months (as of 30 November 2023).

First 90 days

Days	What I'd do	Deliverable
1–30 · Reconcile	Learn how outages, meter events and circuits are recorded and reported. Rebuild one year's SAIDI, SAIFI and CAIDI and reconcile them to the filed figures.	A reconciled baseline with written definitions
31–60 · Diagnose	Interruptions by circuit and cause code, vegetation included; customers interrupted repeatedly (CEMI). Restoration timed from meter “last gasp”: 99.8% of Dominion's meters were smart meters in 2025.	Tested views and a data dictionary
61–90 · Prioritise, track	Rank feeders by customer-minutes; track mainfeeder hardening against similar untreated feeders and weather, building on Dominion's published before/after measure.	**A Power BI programme-tracking report on a scheduled refresh**

Logistics

- Time, place and format: [you: …]
- Panel names and roles: [you: …]
- Work authorisation, if asked: [you: …]
- Salary, if asked: give a researched range only if pressed. [you: your range]

How it's built

Built like governed production reporting, and the numbers to know

Public files land in Snowflake as raw text exactly as published (RAW), are typed and cleaned (STG), and become facts, dimensions and views (MART) that Power BI imports. Twelve tests run after each load (T1–T12, all pass). A data dictionary gives every column and measure a business definition, and a decisions log records fourteen choices (D1–D14), such as which storm types count and which peers to use, made before results were seen. The first check was reconciliation: 130.2 and 133.0 rebuilt against 130 and 133 reported; in all, 13 of 13 rebuilt values match their published figures. The county outage, storm and load tables add tests T13–T31 (all pass).

Numbers to know

Number	What it is	Source
130 → 133	Reported 3-year SAIDI ex-MED, 2022–24 → 2023–25 (minutes)	Annual Report p. 10
130.2 → 133.0	The same, rebuilt from EIA-861	EIA-861 / PUDL
136.0 → 144.4	SAIDI ex-MED, 2022 → 2025, the swap that moves the average	EIA-861
1.199 → 1.320	SAIFI 2022 → 2025 (+10%); 2025 the highest of 2013–2025	EIA-861
113.5 → 109.4	CAIDI 2022 → 2025, minutes per interruption (−4%)	EIA-861
1.21 · 4th of 12	SAIFI 2021–25 vs peers (median 0.96)	EIA-861 / PUDL
110 · 6th of 12	CAIDI 2021–25 vs peers (median 105)	EIA-861 / PUDL
11.9×	Outage customer-hours per 1,000 customers, storm-report vs calm county-days, 2015–2025 (95% CI 7.9–16.9)	EAGLE-I, NOAA
464 → 791	Storm county-days, 2024 → 2025	NOAA Storm Events
r = 0.70	Storm county-days vs SAIDI ex-MED, 2015–2024, n = 10	NOAA, EIA-861
306.7 vs 144.4	2025 SAIDI with and without major event days (53% MED)	EIA-861
+39–41%	Load, hours ending 1–5 a.m., 2019 → 2025 (1–5 p.m. about +27%)	EIA-930
+26% · +52%	Highest and lowest hour of the year, 2019 → 2025	EIA-930
2019, 2022, 2025	Years winter set the yearly peak (2026 to 29 Oct a tie)	EIA-930
2,675,735	Dominion Energy Virginia customers, 2025	EIA-861

One SAIDI minute across 2,675,735 customers is about 2.7 million customer-minutes.

Appendix A

Definitions, and what public data cannot show

Term	Meaning
SAIDI	Outage minutes per customer per year. SAIDI = SAIFI × CAIDI. Dominion 2025: 1.320 × 109.4 = 144.4 minutes.
SAIFI	Interruptions per customer per year: how often.
CAIDI	Minutes per interruption: how long.
Major event days	Days set aside so one extreme storm does not swamp the trend. IEEE 1366 marks a day when its SAIDI exceeds a threshold set from the previous five years of daily SAIDI (the "2.5 beta" method). Dominion reports under its own rule.
ex-MED	Excluding major event days: the headline series in this brief.
CEMI	Customers experiencing multiple interruptions: a count from outage records.
Log split	ln(SAIDI ratio) = ln(SAIFI ratio) + ln(CAIDI ratio); the frequency share is ln(SAIFI ratio) ÷ ln(SAIDI ratio). 2023 → 2025: 0.125 ÷ 0.158 = 79%.

What public data cannot show, and what would answer it

Public data limit	What would answer it (Dominion data, by type)
One number per utility per year (EIA-861)	Outage records by event, circuit and cause
Storms counted by county, not by line	Outage records with cause codes and circuit location
Methods differ between utilities	Dominion's daily SAIDI and its filing method
The DOM zone is wider than the retail territory	Meter interval data by circuit
No customer identities: load growth cannot be attributed	Customer class and large-load records

Appendix B

The twelve-peer table, sources and tests

Utility	Major-event method	SAIFI 2021–25	CAIDI 2021–25	SAIDI 2023–25
Duke Carolinas (NC)	IEEE 1366	1.25	139	179
Georgia Power	Mixed	1.24	114	141
Duke Progress (NC)	IEEE 1366	1.23	114	144
Dominion Energy Virginia	Own method	1.21	110	133
Dominion Energy SC	IEEE 1366	1.14	72	84
Duke Energy Florida	Mixed	0.99	87	83
Alabama Power	Own method	0.93	125	110
PECO	IEEE 1366	0.74	99	66
BGE	IEEE 1366	0.71	99	64
PPL Electric	IEEE 1366	0.69	137	99
Pepco (MD)	IEEE 1366	0.57	90	48
FPL	IEEE 1366	0.49	96	47

EIA-861 / PUDL, excluding major event days; means of annual values; sorted by SAIFI. Peers chosen by EIA ID before results.

Sources

Source	Publisher	Years	Rows
EIA-861 Reliability, 2025 final	EIA	2025	977
PUDL EIA-861 reliability	Catalyst Cooperative	2013–2024	24,358
EIA-861 Service Territory	EIA	2025	11,803
NOAA Storm Events	NOAA NCEI	2013–2025	831,288
EAGLE-I county outages (Virginia)	ORNL for DOE	Nov 2014–2025	11,608,442
EIA-930 hourly demand	EIA	Jul 2018–Oct 2026	6,077,232

Tests after each load (T1–T12, all pass): no duplicate keys; SAIDI = SAIFI × CAIDI within 2%; Dominion 2025 equals EIA's published 306.7, 144.4 and 2,675,735 customers; every county mapped; full years; no duplicate hours. **T13–T31** check the county outage, storm and load tables (all pass).